

Master Team Project WS2024

FindFul

Team 1

Member Name	Role
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Milestone 2

27/11/2024

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1. Functional Requirements - prioritized:

Priority 1:

User registration and authentication

1. The system shall allow students and landlords to create an account.
2. The system shall allow users to log into their accounts using a unique email and a password combination and upon successful login, the user should be redirected to the homepage.
3. The system must verify the student's enrollment status at the time of registration by confirming that the student email ends with the domain "@informatik.hs-fulda.de," and if the email matches the domain, the system shall confirm the student to be verified.
4. The system shall allow registered users to log out and upon successful logout, the system shall be redirected to the login page.

Profile Management

5. The system shall allow registered users to manage their profiles, including personal information, contact details, and profile picture.

Manage Listings

6. The system shall allow the landlords to post detailed accommodation listings that include essential information such as:
 - Property description
 - Location
 - Rent
 - Amenities
 - Availability
7. The system shall allow landlords to include multiple media files, e.g., images for visual clarity.
8. The system shall allow landlords to edit or remove their accommodation listings.

Moderator Functions

10. The system shall allow the Moderator to approve or reject listings based on their content. Listings will only be visible to students after receiving moderator approval.
11. The system shall allow the Moderator to delete users from the system, who violate the platform's terms of service.

Search & Filtering

12. The system shall allow Users (unregistered and registered) to perform searches for listings using street information such as address, and postal code.
13. The system shall provide filters for searching listings, including:
 - Size of the apartment.
 - Type of the apartment i.e. single, shared etc.
 - Maximum and minimum price range.
 - Number of rooms.

Accommodation Listings

15. The system shall allow users to click on a listing to view detailed information including description, amenities, photos, location on a map, rent breakdown, and availability.
16. The system shall allow users to apply to a listing from the listing detail page.

Group Application for Shared Accommodation

18. The system shall allow students to form a group by adding up to 6 other students in the system.
 - 18.1. The system shall allow students to accept or reject group invitations.
19. The system shall allow students to apply as a group to any rental accommodation that has available rooms for students in a group.

Communication

20. The system shall allow students to apply to an accommodation, in group or individually, by messaging landlords using the in-app messaging app.

21. The system shall have a “My Messages” section for both student and landlord to manage chats.

Priority 2:

Accommodation Listings

16. The system shall suggest additional listings that match the user’s previous search preferences or saved listings.
17. The system shall allow users to compare multiple listings side-by-side allowing them to easily compare prices, discounts, and amenities.
18. The system shall allow users to view the distance of the accommodation from important Landmarks including the city center, Fulda University of Applied Sciences, clinics, grocery stores, etc., on a map.

Profile management

9. The system shall allow students to create their personalized profile cards with information including their hobbies, interests, and characteristics, allowing landlords to view them when the student applies for a listing.

Managing Listings

22. The system must allow students to add their preferred listings to favorites for later access.

Priority 3:

Search & Filtering

14. The system shall display a map showing the locations of all listings that match the user's selected filters.
15. The system shall provide a "save search" feature that allows users to save their search criteria for future use.

Managing Listings

24. The system shall allow students to rate a listing/landlord on a scale of 1-5.
25. The system shall provide a real-time notification feature so the student is notified of any new relevant listings.
26. The system shall provide a dashboard for the landlord to view main statistics for the listing which include the total number of views in a day/week on the listing.
27. The system shall allow registered users to report:
 - 27.1. A listing if they see something inappropriate in the post.
 - 27.2. A landlord if they break the rules.
 - 27.3. A student if they break the rules.
28. The system shall notify the Moderator of the reported listings, landlords, and students to take an action.
29. The system shall provide an interactive 3D view of the accommodation, allowing users to rotate, zoom, and pan a 3D render.

Communication

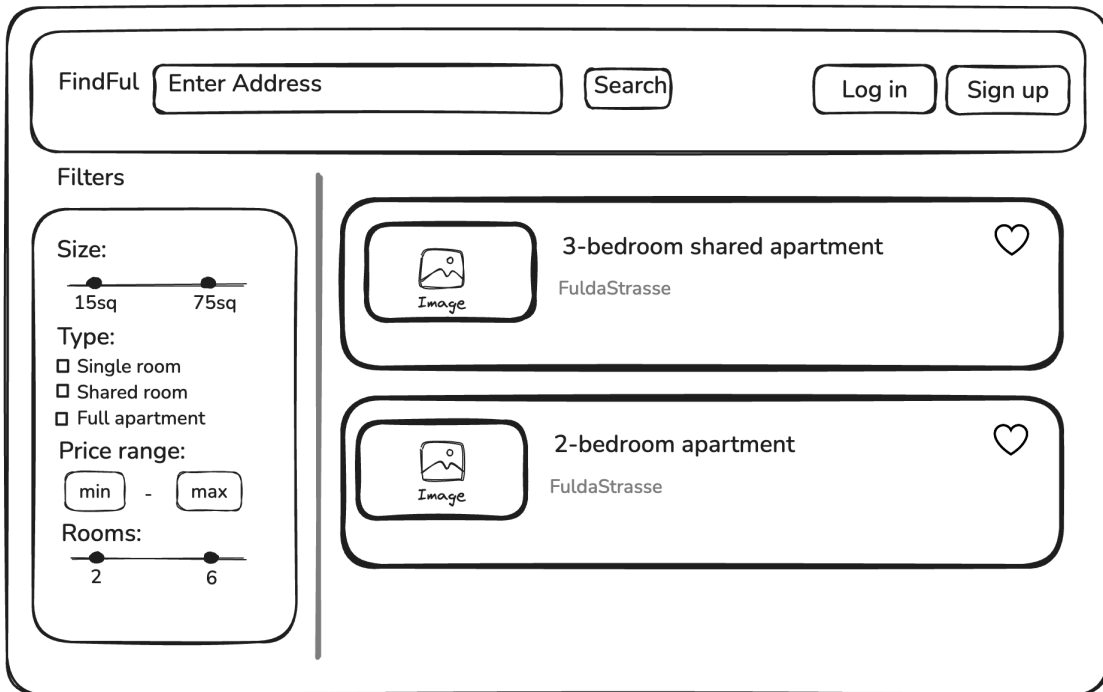
30. The system shall allow in-app messaging between two students.
31. The system shall allow users to share files, images, and videos through chat.
32. The system shall allow users to move chats from the list of active chats to an archive.

2. List of main data items and entities (expand as necessary):

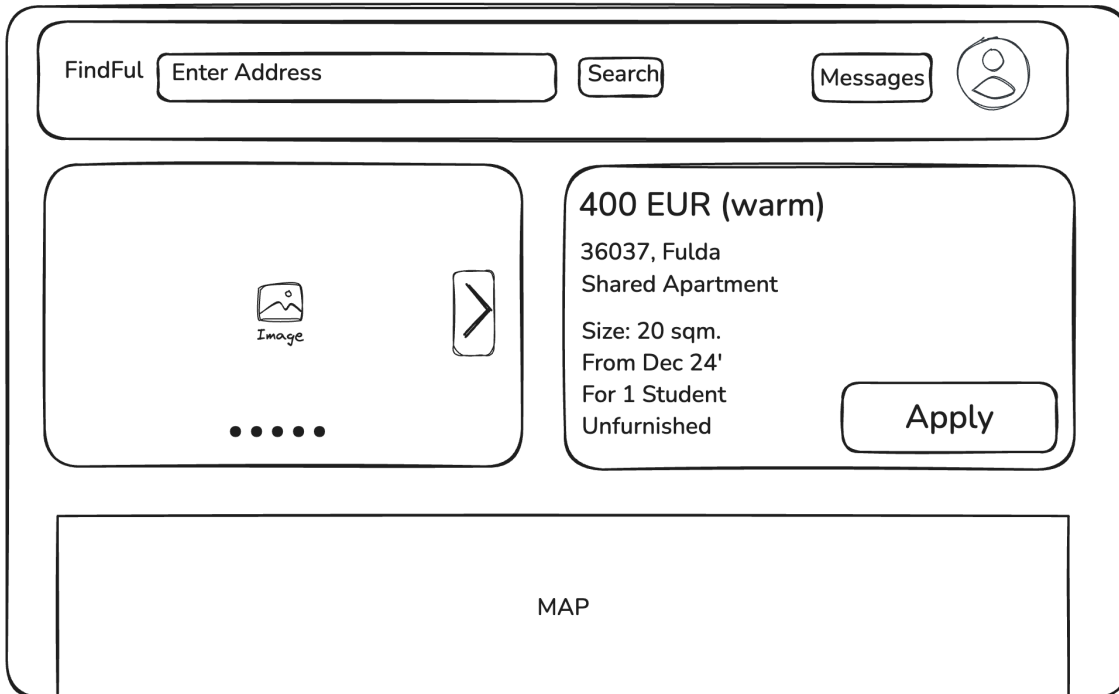
1. **Student** represents a verified student of the Fulda University of Applied Sciences. A student must complete email verification (i.e register with a hs-fulde.de email) to access the platform.
2. **Landlord** is any individual or organization offering rental properties on the platform. There is no email verification for a landlord and they can use any email address to register.
3. **Guest** is an unauthenticated visitor with limited platform access. They can browse public listings, view basic property information, and see location details, but cannot contact landlords or access full listing features.
4. **Moderator** is a system maintainer with comprehensive moderation powers. They review all new property listings and associated media, with the authority to approve or reject content based on platform guidelines.
5. **Real estate** is a representation of physical property available for rental, specifically targeting student accommodation in Fulda.
6. **Listing** is the central data element representing a property available for rent. It contains all property details including title, description, monthly rent, and property type.
7. **Student Group** is when a single student creates a group with a few other students to apply to an accommodation together.
8. **Media Item** is visual or image content associated with a rental listing, serving as a critical representation of the property's physical appearance and characteristics.

3. NEW: UI Mockups and Storyboards (high level only):

Use Case 1: Search for Housing - Tim (Guest)



Use Case 2: View Selected Housing - Ram (Student)



Use Case 3: Connect with Landlords - Ram (Student)

FindFul

Search



Mr. Muller

Ms. Sandy

Mr. Tim

3 Room shared Bedroom Apartment....

Hallo, I like the apartment location.....

Sounds good!

Use Case 4.1: Apply as a Group - Ram (Student)

FindFul

Search

Messages



Apply for

☐ Apply as an individual

☒ Apply as a group

Select a group



Message:

Apply

Use Case 4.2: Apply as a Group - Create Group

FindFul

Search

Messages



Apply for

☐ Apply as an individual

Create a new group

Members:

Add user by username:

Create

Apply

Use Case 5.1: Add Listing of New Real Estate - Thomas (Landlord)

FindFul

Search

msg

+



Filters

Size:

15sq 75sq

Type:
☐ Single room
☐ Shared room
☐ Full apartment

Price range:
 -

Rooms:

2 6


Image

3-bedroom shared apartment
FuldaStrasse




Image

2-bedroom apartment
FuldaStrasse

Use Case 5.2: Add Listing Form

FindFul

Search

msg

+



Details

Title

Price

 EUR

Size

 Sq m.

Rooms available

Available from

Type of apartment

Choose one

▼

Single room

Single apartment

Shared room

Description

Images



 clear

Location Details

Zip code

Street/ No.
(optional)

Add

Use Case 6.1: Reviewing New Listings - Anna (Moderator)

FindFul



LISTINGS

Review

Approves

Rejected


Image

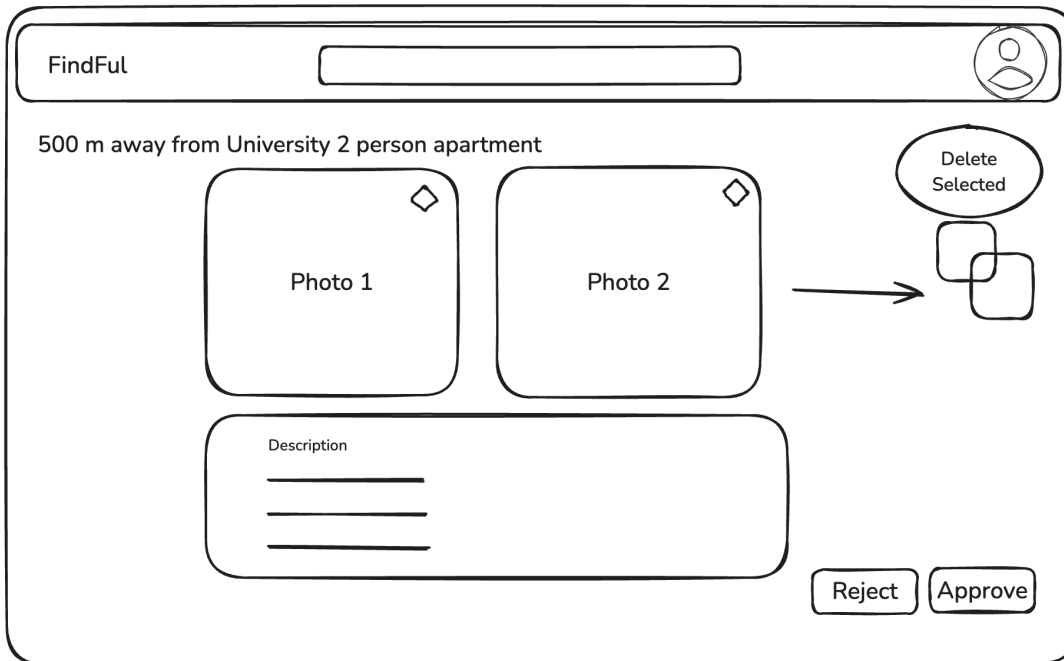
18sqm Apartment near University


Image

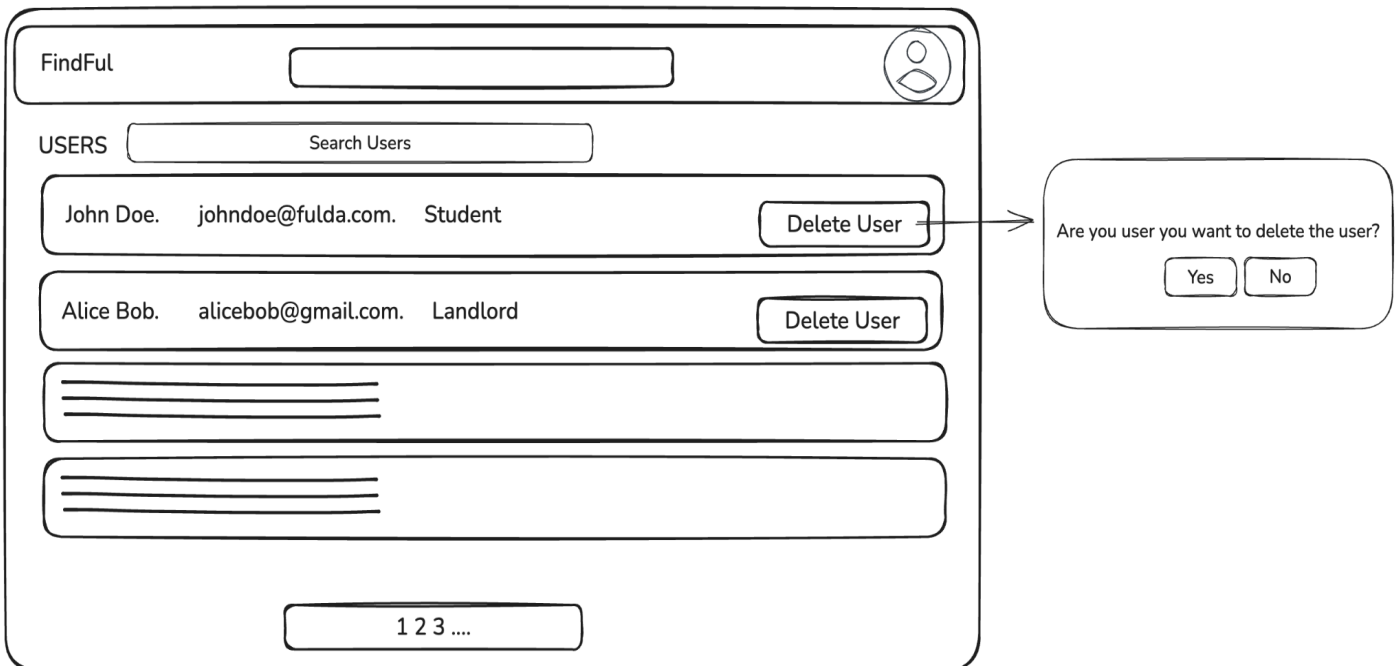
3-person shared flat in Fulda

1 2 3

Use Case 6.2: Delete Inappropriate Listing/photo

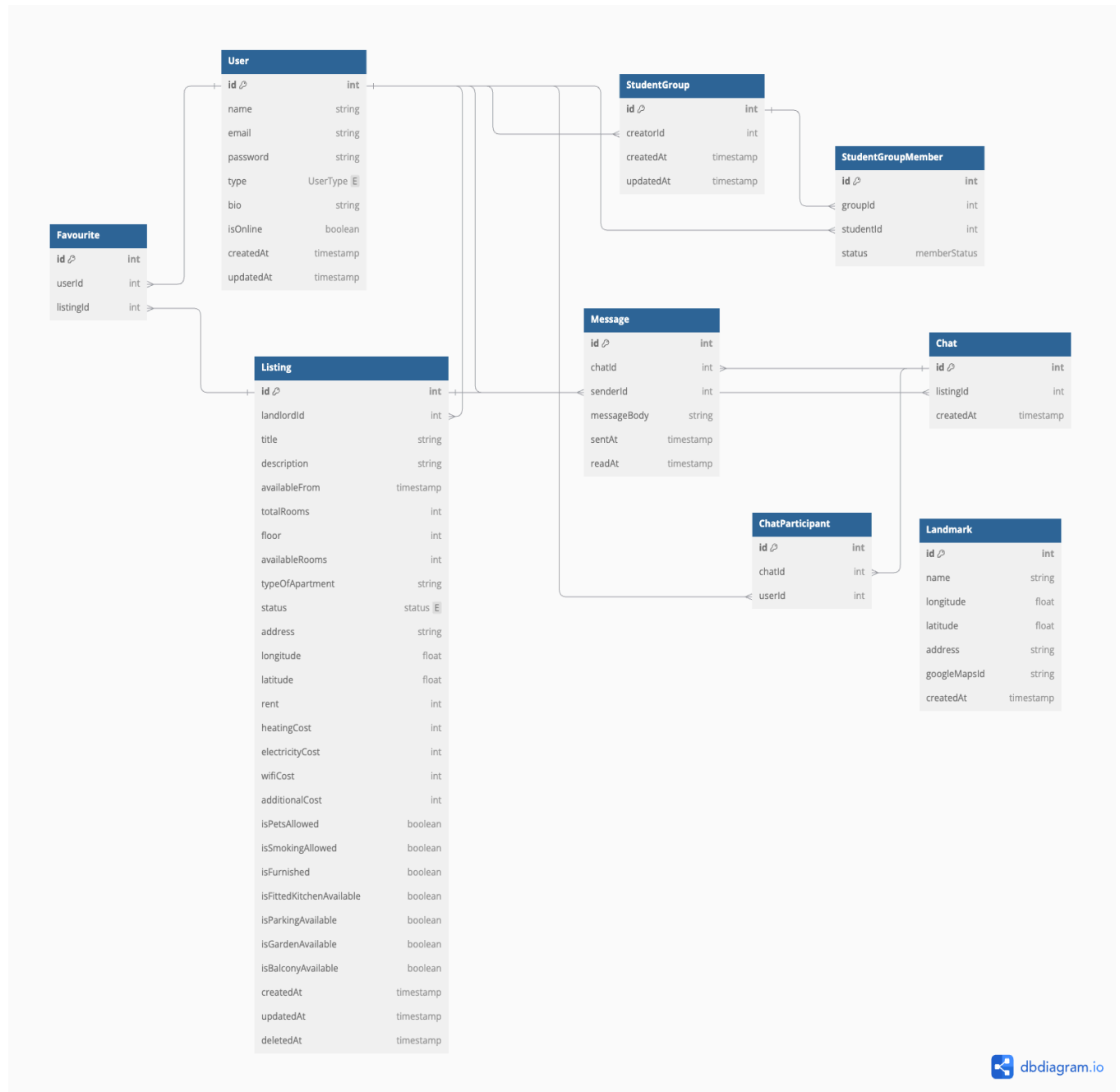


Use Case 6.3: Delete Inappropriate Users



4. NEW High level Architecture, Database Organization:

Database Schema:



The FindFul Database consists of the following main tables:

1. **User:** The User table consists of three types of users (STUDENT, LANDLORD, MODERATOR), along with user email, password (stored as a hashed string), bio, and isOnline.
2. **Listing:** The Listing table is created when a User (LANDLORD) creates a new listing for the moderator to review. It consists of a landlordId (referenced from the Users table) along with other attributes like address, zip code, rent, description, date available from, totalRooms, availableRooms, typeOfApartment, etc. It also consists of some key amenities stored as booleans, like isPetsAllowed, IsKitchenFitted, isFurnished, etc.
3. **StudentGroup:** In order to apply for accommodation as a group, a student can create a group by inviting other students in the system. The table includes a creatorId (referenced from the Users table).
4. **StudentGroupMember:** When a student is sent an invitation to join a group, their record is added to this table with a status (ACCEPTED, PENDING, REJECTED) along with the StudentId (referenced from the Users table) and the groupId (referenced from the StudentGroup table).
5. **Favourite:** Students and Landlords can favorite their desirable listings for the purpose of viewing later. The table stores the listingId (referenced from the listings table) and the userId (referenced from the Users table).
6. **Landmark:** This table has prefilled data of all the nearest landmarks a student would want to have near the apartment, i.e., University, grocery stores, clinics, city center, etc. We store the longitude and latitude for the distance to be calculated later by the Maps API.
7. **Chat:** Whenever a student applies to a listing by sending a message to the landlord, a chat is created with a primary id, and listingId.
8. **ChatParticipants:** For the purpose of one chat having more than two participants, we use this table, which will have the participantId (STUDENT or LANDLORD) and the chatId (referenced from the Chat table).
9. **Messages:** This table stores the message body, along with senderId (foreign to the User table), and chatId (foreign to the Chat table).

Media Storage:

The user and apartment photos will be stored in an S3 bucket in JPEG format. JPEG is the best choice due to its lossy compression, which is optimal for photos. It provides a good balance between quality and file size. Since being able to view apartment photos is crucial for this application, they need to be of good quality, and JPEG should fulfill that requirement effectively. Additionally, JPEG files are smaller in size compared to PNGs.

The photos can be stored in the S3 bucket in respective folders according to the database tables.

1. User Photos:

- a. **Storage Location:** users/
- b. **File Name Convention:** User photos will be stored in the user folder with the user ID as the filename.
- c. **Example:** users/<user_id>.jpeg

2. Apartment Listing photos:

- a. **Storage Location:** listings/
- b. **File Name Convention:** Photos of listings will be stored under the listings folder. This folder will contain subfolders named with the GUID of the listing from the database table, and multiple photos of the listing will be stored within the corresponding subfolder.
- c. **Example:** listings/<listing_id>/<photo_name>.jpeg

3. Thumbnails for Listing:

- a. **Storage Location:** listings/<listing_id>/thumbnails/
- b. **File Name Convention:** The thumbnails for the listing will be compressed and stored in the folder associated with the listing, using the listing_id under the /thumbnails folder, so they can be retrieved easily.
- c. **Example:** listings/<listing_id>/thumbnails/<photo_name>.jpeg

Search/filter architecture and implementation:

In FindFul, filtering is used to refine a list of listings that is presented to the user, by applying filters chosen by the user. The filtering is implemented with vanilla MySQL. Filtering with MySQL queries instead of JavaScript or TypeScript has the advantage of requiring less data to be stored. It is faster since the data needs to be fetched only once. All filters that can be applied by the user are precomposed filters that directly relate to fields in the database.

API Structure:

We will implement custom APIs to support the application's functionality, providing endpoints for user management (/users), real-estate listings (/listings), search and filtering (/search), and group management (/groups). The /users endpoint will support GET, POST, PUT, and DELETE requests, enabling functionalities such as user registration, authentication, authorization, and updating or deleting user information.

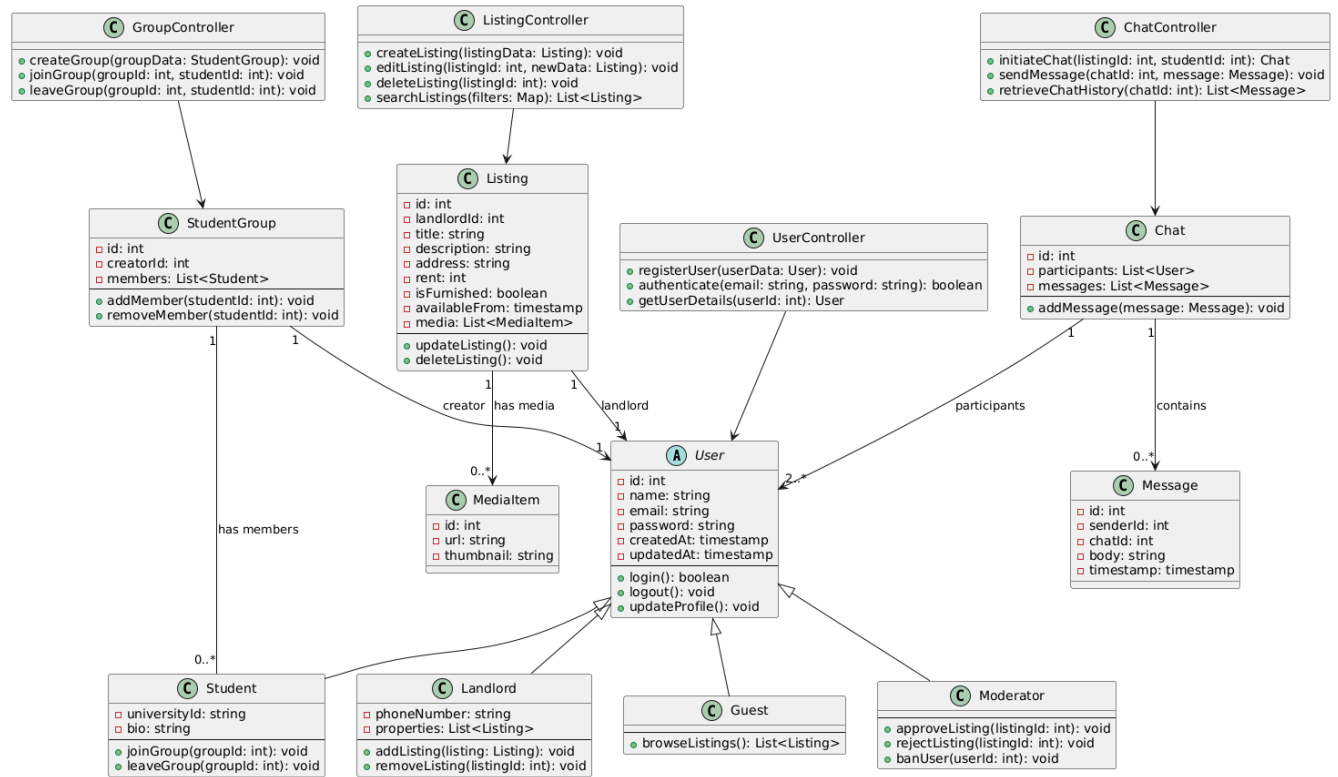
The /listings endpoint will allow for adding new apartments, updating apartment details, retrieving apartment information, and deleting listings.

The /search endpoint will handle search requests and apply filters to provide appropriate listing responses.

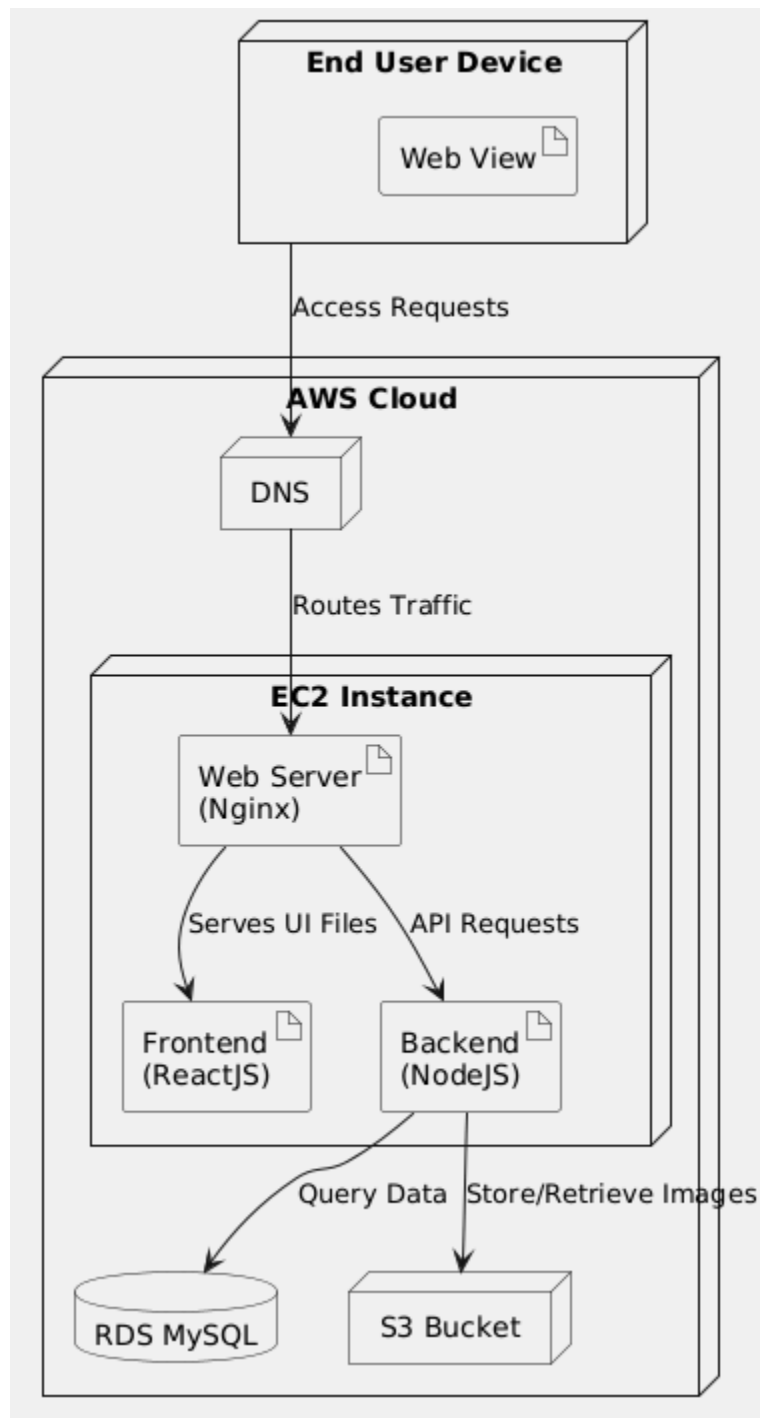
Additionally, the /groups endpoint will manage the creation, deletion, and assignment of groups and their members.

5. High Level UML Diagrams:

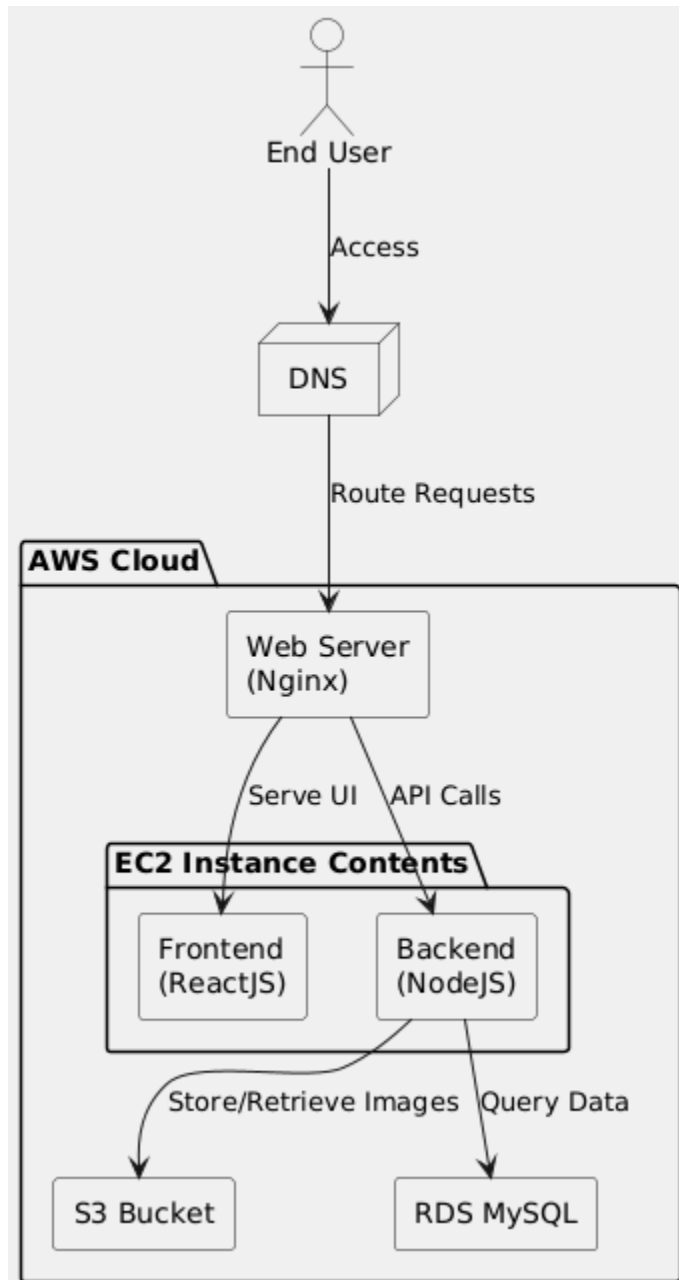
a) UML class diagram



b) UML Deployment diagram



c) Component diagram



6. Identify actual key risks for your project at this time:

1. Skill risks:

1.1 Risk:

Our team does not have a team member with a strong background in frontend development. This lack of skills poses a risk for our project since the frontend is a significant part of it.

How we plan to overcome it:

To address this, we have chosen a frontend lead who plans to continue researching our chosen frameworks, modern frontend methodologies, and best practices. The other team members will assist the frontend lead with their prior knowledge or newly acquired knowledge when needed.

1.2 Risk:

The technical skill levels of our team members vary, as some team members have extensive prior knowledge of software development while others do not. This may challenge the task distribution and communication among other team members.

How we plan to overcome it:

The task distribution is done with the skill levels of the team members in mind. If more knowledge is needed to complete a task, the team member will research it and also receive support from others. The team regularly discusses the task distribution, and if beneficial, task solutions are presented.

2. Schedule risks:

2.1 Risk:

Our weekly internal team meeting on campus takes place on Monday morning. If someone is unable to participate in it, we unfortunately do not have another time to meet up in person.

How we plan to overcome it:

So far, it has proven to be a good alternative to meet up online in the evenings. These online meetings are also used to explain new technologies to the team. If an in-person meeting is unavoidable and someone cannot attend, the team will hold the meeting and update the absent team members with a brief protocol of the meeting.

2.2 Risk:

Another scheduling risk is the amount of time we have during the semester and the examination phase. Some team members are taking multiple exams in February, which results in less time for the project during that period.

How we plan to overcome it:

To achieve good results for both the FindFul project and the exams, we will adjust our project timelines to complete the most important and critical features at the start, leaving us with smaller features close to the examination phase. We will also keep the task distribution flexible, meaning team members with other educational commitments in a week can contribute less during that week and make up for it in the coming weeks.

3. Legal Risk

3.1 Risk

The FindFul platform uses images to present the different accommodations with listings to guests and students. This requires a lot of images that must be free to use.

How we plan to overcome it:

The team intends to use copyright-free images from websites like Pixabay, Pexels, or Free Images. These websites offer a wide range of images that are free to use for non-commercial projects. For each image, the license will be checked for a correct usage.

7. Project management:

For our team project management, we are using Trello. With Trello, our team is able to create different tasks, assign them to team members and Trello helps the team to get an overview of the project.

We divided our Trello board into six sections. The first section is “To-Do,” in which we collect future tasks. Once a team member actively works on a task, the task is moved to the section “Doing.” If the team member has finished the task, they put it in the section “Review” for other team members to review it. Tasks that are ready for production are placed in “Done.” In the section “Resources,” we collect all outside resources that we are using to be able to access them quickly. Besides links to our milestone documents or our GitHub rules, we collect links to tutorials that concern every team member. Lastly, we use the section “Weekly Meeting Notes,” where we put notes about our meetings and the To-Dos that developed from them.

Trello automatically adds an ID to the tasks. We use this ID in our Git commit messages for a better overview of what the code change was about.